

Democratising Personalised STEAM Education: A Generative AI-Driven Intelligent Tutoring System Aligned with Standardised Pedagogical Frameworks in Mexico

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ABSTRACT.

This paper presents the design, and implementation, of an Adaptive STEAM Virtual Tutor — an Intelligent Tutoring System (ITS) built on Google Colab and powered by the Gemini Flash generative model via the Google Gen AI Python SDK. The system automates a three-stage pedagogical pipeline: (1) diagnostic assessment delivered through Google Forms and recorded in Google Sheets, (2) algorithmic performance classification into three levels — Bajo ($\leq 50\%$), Normal ($60\%–75\%$), and Sobresaliente ($80\%–100\%$) — using a custom Python function, and (3) personalised activity generation by Gemini Flash, which produces level-differentiated reinforcement exercises and teacher recommendations via a structured prompt.

The system's evaluation consisted of an initial pilot in Mathematics and Complexity Sciences at CBTIS 131 (a public federal upper secondary school- equivalent to Sixth Form in the United Kingdom) with a cohort of fewer than 20 students over two weeks. Results validated against independent CENEVAL-aligned post-tests show a 20% average group improvement (6.5/10 to 8.5/10), individual gains from 50% to 100% in the primary case study, and a reduction in teacher profiling time from approximately one week to under five minutes. The system is distinguished by its zero-cost deployment on pre-existing Google Workspace infrastructure, its alignment with both the CENEVAL framework and the DGETI/SEP Marco Curricular Común, and its affective pedagogical design targeting math anxiety reduction

Keywords: Intelligent Tutoring Systems, Generative AI, Gemini Flash, Google Colab, STEAM Education, CENEVAL, Mexico, Adaptive Learning, Bachillerato, MCC DGETI.

1 INTRODUCTION

The global education landscape faces a persistent and widening challenge: ensuring that students develop genuine proficiency in STEAM disciplines—Science, Technology, Engineering, Arts, and Mathematics—at a time when classroom instruction increasingly struggles to meet individual learners where they are. As Anderson, Boyle, and Reiser [1] observed four decades ago, Intelligent Tutoring Systems are "programs that simulate an understanding of the domain they teach and can respond specifically to the student's problem-solving strategies." That foundational insight has lost none of its relevance; if anything, the emergence of generative artificial intelligence has made it more urgent and more achievable than ever before.

In Mexico's upper secondary education—*Educación Media Superior*—these pressures are especially acute. Students at public technical institutions such as *Centro de Bachillerato Tecnológico*

Industrial y de Servicios Núm. 131 (CBTIS 131) regularly encounter conceptual thresholds—self-similarity in fractal geometry, non-linear dynamics in Complexity Sciences—that a single 50-minute session, shared among an entire heterogeneous cohort, cannot adequately address. National curricular frameworks, from the *Marco Curricular Común* (MCC) of the DGETI/SEP system to the assessment guidelines of CENEVAL, demand competency-based learning and higher-order thinking [2,3], yet the resources available to teachers in these settings remain severely constrained.

Research spanning several decades offers a compelling response to this gap. VanLehn [4] demonstrated that well-designed computer-based tutors can approach the effectiveness of one-on-one human instruction in specific educational contexts, while Graesser et al. [5] showed, through AutoTutor, the particular power of natural language dialogue in fostering deeper conceptual understanding. Meanwhile, Woolf [6] emphasized the importance of adaptive learning strategies that respond dynamically to each student's performance, pace, and cognitive needs. Together, these findings point toward a clear design agenda: personalized, conversational, and adaptively responsive tutoring at scale.

Recent advances in large language models have brought that agenda within reach for everyday educational contexts. Google's Gemini Flash model, accessible through open programming interfaces, offers sophisticated natural language understanding and real-time content generation [7]—capabilities that align closely with Krashen's [8] Input Hypothesis, which stresses the importance of comprehensible, context-sensitive interaction for meaningful learning. By embedding such a model within accessible, zero-cost infrastructure, it becomes possible to build an intelligent tutor that is not only pedagogically sound but also practically deployable in under-resourced schools.

This article presents the design and implementation of the *Adaptive STEAM Virtual Tutor*, an Intelligent Tutoring System developed on Google Colab and powered by the Gemini Flash generative model via the Google Gen AI Python SDK. Piloted at CBTIS 131 in San Luis Potosí, the system integrates adaptive questioning, formative feedback, automated assessment generation, while remaining fully aligned with the CENEVAL national evaluation framework and the MCC competency standards of the DGETI/SEP system. The study describes the system's architecture, its pedagogical foundations, and the implementation process, demonstrating how state-of-the-art generative AI can serve as the engine of an effective, scalable, and curriculum-aware educational tool for students in *Educación Media Super*

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2 RELATED WORK AND CONTRIBUTIONS

Contemporary ITS solutions span three paradigms as shown in figure 1: constraint-based systems (Carnegie Learning MATHia) use Bayesian Knowledge Tracing (BKT) and expert-encoded rule sets to deliver deterministic, step-level feedback; over two decades of empirical evidence confirm learning gains comparable to one-on-one human tutoring [9, 10], though their construction requires extensive domain engineering by subject-matter experts [11].

Data-driven systems such as ASSISTments infer student knowledge states from large interaction logs rather than hand-coded rules [12]; advances in deep learning — from Deep Knowledge Tracing [13] to Transformer-based architectures [14] — have substantially improved predictive accuracy, while the platform's open dataset infrastructure has established it as a benchmark for educational data mining research.

LLM-based tutors such as Khan Academy's Khanmigo and GPT-4-powered systems represent the most recent paradigm, offering natural-language dialogue and broad domain coverage; a systematic review of experimental studies reports improvements in academic performance, affective-motivational states, and higher-order thinking [15], yet hallucination — the generation of plausible but factually incorrect feedback — remains a critical reliability concern documented across mathematics and science domains [16, 17].

Figure 1. Intelligent Tutoring Sysyem paradigms illustrated with everyday analogies



As illustrated in Figure 1, the three paradigms differ not only in their underlying technical architecture but also in where they place the burden of pedagogical knowledge: constraint-based systems locate it in the expert who encodes the rules; data-driven systems distribute it across the historical behaviour of millions of students; and LLM-based systems embed it implicitly within the parameters of a pre-trained language model. This distinction has direct consequences for scalability, reliability, and contextual validity. While each paradigm offers a distinct combination of strengths — diagnostic precision, data-driven adaptability, and conversational fluency, respectively — none addresses simultaneously the cost

barriers of under-resourced public education systems, the absence of alignment to non-Anglo-American curricula, and the affective dimensions of learning in high-stakes secondary contexts. The proposed system is designed precisely at this intersection: it inherits the conversational capability of LLM-based tutors while constraining generative output to formally specified learning objectives drawn from the CENEVAL item criteria and the DGETI/SEP MCCEMS competency framework, functioning as a structural hallucination-mitigation strategy absent from all three existing paradigms as currently deployed.

Against this backdrop, our system is differentiated along three axes:

- **Zero-cost deployment:** The entire stack — Google Forms, Google Sheets, Google Colab, and the Gemini Flash API — is available at no cost to Mexican public schools already operating within the Google Workspace for Education ecosystem, unlike licensed ITS platforms.
- **Mexican curricular alignment:** Existing ITS solutions predominantly target Anglo-American curricula. Our system is explicitly designed under CENEVAL criteria for formally structured test items and the DGETI/SEP Marco Curricular Común (MCCEMS), providing construct validity within the Mexican formal assessment context through a `definer_nivel()` function that maps each generated activity to a specific competency level prior to delivery. Critically, this curricular anchoring also serves as a hallucination-mitigation strategy: by constraining generative output to formally specified learning objectives and competency levels, the system reduces the risk of off-domain or pedagogically unsound content generation [16].
- **Affective pedagogy:** The system's prompt engineering incorporates an empathetic teacher persona and real-world analogies to reduce the affective filter [1] — a documented barrier to STEAM learning in high-stakes Mexican secondary contexts.

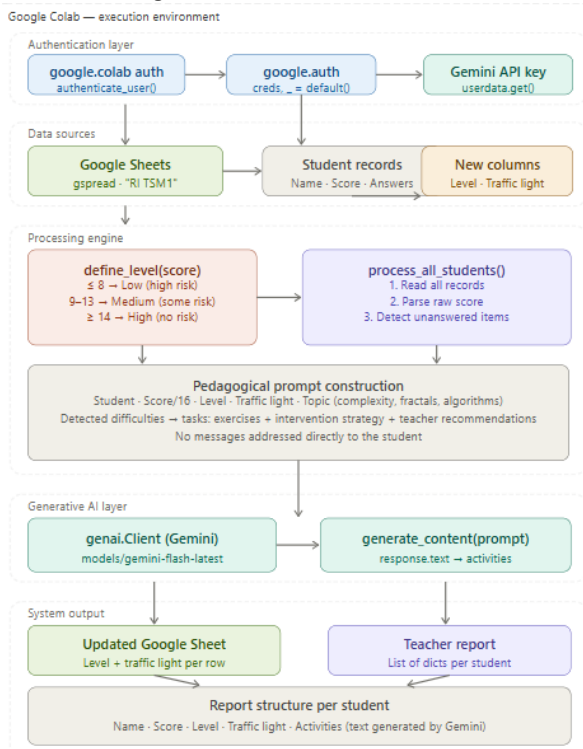
3 SYSTEM ARCHITECTURE AND TECHNICAL IMPLEMENTATION

The system is implemented as a Google Colab notebook using Python. The full technical stack comprises: the `google-generativeai` and `gsread` Python libraries, the Google Colab authentication module, the Gemini Flash generative model (`models/gemini-flash-latest`), and Google Sheets as the persistent data layer.

As shown in Figure 2, the pipeline consists of five layers: (1) an *authentication layer* using `google.colab.auth` and `google.auth` to obtain credentials for Google Sheets and the Gemini API; (2) a *data sources layer* that reads student records (name, score, answers) from the Google Sheets spreadsheet "RI TSM1" via `gsread`; (3) a *processing engine* that applies the `define_level(score)` function to classify students into three risk levels (Low: ≥ 8 , Medium: 9–13, High: ≥ 14) and the `process_all_students()` function to parse scores and detect unanswered items; (4) a *pedagogical prompt construction layer* that assembles personalized prompts encoding student score, risk level, topic, detected difficulties, and teacher recommendations; and (5) a *generative AI layer* using `genai.Client` (Gemini Flash) and `generate_content(prompt)` to produce individualized activity sets. System outputs include an updated Google Sheet with level and traffic-light indicators per row, and a teacher report as a

list of dictionaries per student, structured by name, score, level, traffic light, and AI-generated activities.

Figure 2. SYSTEM ARCHITECTURE



3.1 Authentication Layer

The notebook begins by installing and importing the required libraries and authenticating against Google services:

Figure 3. Python authentication setup: library imports, Gemini API key retrieval from Colab secrets, and OAuth2-bases Google Sheets authorization

```
!pip install --upgrade -q google-generativeai gspread

from google.colab import userdata, auth
from google import genai
import gspread
from google.auth import default

API_KEY = userdata.get('GEMINI_API_KEY')
client = genai.Client(api_key=API_KEY)

auth.authenticate_user()
creds, _ = default()
gc = gspread.authorize(creds)
```

Figure 3 shows the authentication steps. The Gemini API key is stored as a Colab secret (userdata), avoiding hardcoded credentials. Google Sheets authentication uses OAuth2 via the Colab built-in flow, ensuring institutional data remains within the Google Workspace security perimeter.

3.2 Data Source: Google Sheets Integration

Figure 4. Work flow for reading evaluation records from Google Forms into the system

```
SPREADSHEET_NAME = "ED TSM1"
sheet = gc.open(SPREADSHEET_NAME).sheet1
data = sheet.get_all_records()
ultimo_estudiante = data[-1] # most recent submission
```

As shown in Figure 4, student diagnostic data is read directly from a Google Sheets spreadsheet named 'ED TSM1', which collects responses from a Google Forms diagnostic assessment. The system accesses the first sheet and reads all records as a list of dictionaries.

Figure 5. Google Sheets data ingestion: reading students records from the spreadsheet and parsing the diagnostic score

```
valor_sheet = sheet.get_all_records()
ultimo_estudiante = valor_sheet[-1]
puntuacion = ultimo_estudiante['Puntuación (puntuación máxima 8/10)']
puntuacion = int(str(valor_sheet).split()[0])
```

As seen in Figure 5, the diagnostic score is extracted from the column 'Puntuación (puntuación máxima 8/10)', with robust string parsing to handle Google Sheets' varied numeric formats.

3.3 Algorithmic Performance Classification

A custom Python function classifies each student's score (out of 8) into one of three performance levels ("Bajo" "Normal" "Sobresaliente") as shown in Figure 6.

Figure 6. Python implementation of definir_nivel

```
def definir_nivel(p):
    if p <= 4:
        return "Bajo" # Basic: guided exercises
    elif 5 <= p <= 6:
        return "Normal" # Intermediate: applied practice
    else:
        return "Sobresaliente" # Advanced: deepening tasks
```

The level determination algorithm — Bajo ($\leq 4/8$, 0–50%), Normal (5–6/8, 62.5–75%), Sobresaliente (7–8/8, 87.5–100%) — maps directly to differentiated instructional strategies, ensuring the AI-generated content is calibrated to the student's specific learning needs rather than delivering uniform content.

For students at a low achievement level, the system generates lessons comprising guided exercises and explanations tailored to the student's content. For intermediate-level students, an application exercises are provided, whereas for high-achieving students, exercises are generated, requiring specific research or more complex problem solving.

3.4 Personalised Lesson Generation via Gemini Flash

The core of the system is a structured prompt passed to the Gemini Flash model (models/gemini-flash-latest), which generates level-differentiated reinforcement activities and teacher recommendations. The prompt shown in figure 7 incorporates the student's name, diagnostic score, performance level, and explicit pedagogical directives for each level

The prompt instructs Gemini Flash to adopt the persona of an expert teacher in Mexican upper secondary education, generating content appropriate to the student's diagnosed performance level.

Figure 7. Structured pedagogical prompts, that produce level-diferenriated reinforcemente activitties and teacher recommendations,

```
prompt = f"""
Eres un docente experto en educación media superior.

Alumno: {nombre}
Puntuación: {puntuacion}/8
Nivel de desempeño: {nivel_actual}

Diseña actividades de refuerzo acordes al nivel:
- Bajo: ejercicios guiados
- Normal: aplicación
- Sobresaliente: profundización

Incluye recomendaciones para el docente.
"""

response = client.models.generate_content(
    model="models/gemini-flash-latest",
    contents=prompt
)
```

The explicit level-strategy mapping (guided exercises for Bajo, applied practice for Normal, deepening tasks for Sobresaliente) ensures the AI's output is pedagogically differentiated rather than generic. The inclusion of teacher recommendations within the same output serves the dual function of supporting the student and reducing the teacher's planning burden.

3.5 System Output

The function returns shown in figure 8 a structured output combining the student's profile data and the AI-generated activities:

Figure 8. Structured return value of the tutoring piprlinr, formatted for direct sharing with both students and teacher?

```
return f"""
Alumno: {nombre}
Puntuación: {puntuacion}/8
Nivel: {nivel_actual}

--- ACTIVIDADES SUGERIDAS ---
{response.text}
"""
```

This output is human-readable and designed to be shared directly with both the student (personalised reinforcement plan) and the

teacher (recommendations for follow-up instruction), making the system immediately actionable within the classroom workflow.

4 METHODOLOGY AND PILOT IMPLEMENTATION

4.1 Pilot Design

The pilot was conducted at CBTIS 131, San Luis Potosí, México, over two weeks with a cohort of fewer than 20 students enrolled in Mathematics and Complexity Sciences (Pensamiento matemático / sistemas complejos) at upper secondary level. The study followed a pre-post intervention design with three measurement points: (1) baseline diagnostic via Google Forms (scores recorded in Google Sheets 'ED TSM1'), (2) AI-generated personalised activity delivery, and (3) independent post-intervention assessment using CENEVAL-aligned instruments.

A critical methodological decision was the strict separation between the instructional engine (Gemini Flash-generated activities) and the evaluative framework (independent CENEVAL-aligned post-tests). This separation ensures that measured learning gains reflect genuine acquisition rather than familiarity with the AI's output format.

4.2 Case Study: Wendy

To illustrate the system's operation, we present the case of one participant — anonymised as *Wendy* — whose Google Forms diagnostic score of 4/8 classified her as *Bajo*, triggering the guided-exercise pedagogical strategy. The system identified a conceptual gap between symbolic procedural fluency and applied understanding of non-linear systems.

The Gemini Flash-generated output for Wendy's level adopted an empathetic teacher persona, using urban traffic flow as an analogy for non-linear system behaviour — a real-world contextualisation strategy targeting the affective filter. On the independent CENEVAL-aligned post-test, Wendy achieved 100% (16/16), compared to her baseline of 50% (4/8).

5 ANALYSIS OF RESULTS

Table 1. SUMMARY PILOT RESULTS

Evaluation Metric	Baseline	Post-Intervention	Net Variance & Pedagogical Impact
Avg. group score — Mathematics	6.5 / 10	8.5 / 10	+20% group improvement
Avg. group score — Complexity Sciences	6.2 / 10	8.1 / 10	+19% relative gain
Case study — Wendy (Bajo → Sobresaliente)	4 / 8 (50%)	16 / 16 (100%)	Performance recovery (+50%)
Systemic Efficiency: Teacher profiling time	~1 week (manual)	~5 min (automated)	>99% reduction in profiling time
Early At-Risk Identification (Bajo Level)	0% (reactive)	100% (predictive)	Automated diagnostic triggering

5.1 Controlled Pedagogical Performance and Withdrawal Effects

Average group scores in Mathematics improved from 6.5/10 to 8.5/10 (20% gain) on CENEVAL-aligned instruments; in Complexity Sciences from 6.2/10 to 8.1/10 (19% relative gain). The primary case study demonstrates maximal performance recovery in a single intervention cycle, validating the level-classification-to-prompt-strategy mapping implemented in the `definir_nivel()` function.

5.2 Systemic Efficiency

The automated pipeline — from Google Forms submission to Google Sheets ingestion to Gemini Flash generation — reduced student profiling time from approximately one week (manual assessment, error analysis, and differentiated lesson preparation) to under five minutes per student. This efficiency gain is structurally significant for under-resourced Mexican public institutions where one teacher may serve 40+ students per cohort.

5.3 Early At-Risk Identification

The `definir_nivel()` classification function identifies Bajo-level students ($\leq 4/8$) before disengagement, automatically triggering guided-exercise scaffolding protocols. This predictive capability — absent from standard classroom instruction — enables intervention at the point of first failure rather than after repeated poor performance.

6 DISCUSSION AND LIMITATIONS

The results confirm that effective ITS deployment does not require proprietary software. The system's reliance on the free Google Workspace ecosystem — universally available in Mexican public Bachillerato institutions — means it can be replicated by any teacher with a Google account, a Colab notebook, and a Gemini API key.

Several limitations must be acknowledged: (1) the cohort was small (<20 students) over a short period (2 weeks); (2) the single-institution setting limits generalisability; (3) absence of a control group prevents causal attribution of gains to the AI intervention alone; and (4) long-term retention was not measured. Additionally, the current implementation processes only the most recent Google Forms submission (`data[-1]`), which will require architectural modification for concurrent multi-student deployment in larger cohorts.

7 CONCLUSION AND FUTURE WORK

This paper has presented a fully implemented, zero-cost Intelligent Tutoring System for STEAM education in the Mexican upper secondary federal school, built on Google Colab, Google Sheets, and Gemini Flash. The system's three-stage automated pipeline — Forms-based diagnostic, Python classification, and Gemini-generated personalised activities — produces measurable learning gains validated against CENEVAL-aligned independent assessments, while reducing teacher profiling time by over 99%.

Future work will address the current limitations through: (1) parallel multi-student processing to replace the sequential `data[-1]` approach; (2) expansion to a multi-school cohort across Mexican states; (3) Within-Subject Validation and Cross-Partial Analysis; (4) longitudinal retention measurement; and (5) extension to additional DGETI curriculum subjects. The system offers a replicable blueprint for AI-augmented teaching that any upper

secondary federal school can implement at scale using existing infrastructure. With minimal adjustments to the underlying code and prompt design, this system can be effectively implemented across any educational level, offering a versatile tool for personalized learning environment.

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